



Faculty of Computer Science and Engineering
Ghulam Ishaq Khan Institute of Engineering Sciences and
Technology, Topi

CS231L

Marks: 10

Instructor: Engr. Rumman

Dated: 12/02/2026

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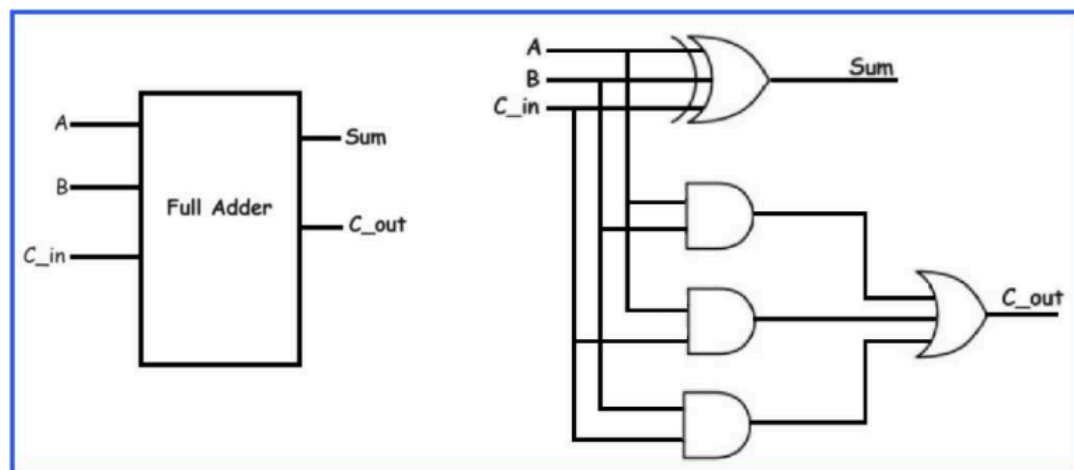
Registration #: 2024206

Lab # 02

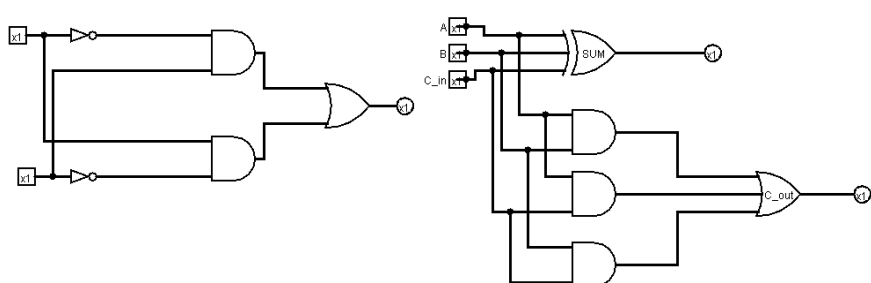
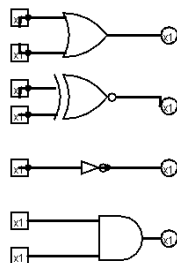
Tasks

Lab Task-1: Implement and Verify All the basic Logic Gates in Logisim.

Lab Task-2: Implement and Verify the Following Full Adder in Logisim



Solution:



Explanation:

1. Basic Logic Gates:

These gates are the fundamental building blocks. OR, XNOR, NOT and AND gates have been tested in Logisim to familiarise how Logisim works and to test basic Logic Gate behaviours.

2. Exclusive-OR (XOR)

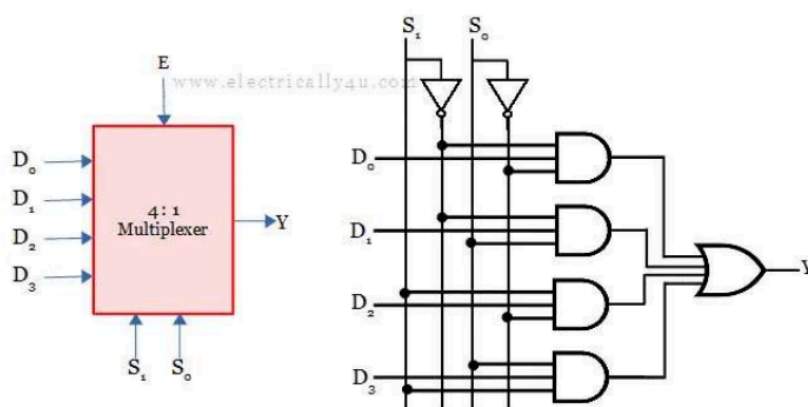
This circuit uses a combination of NOT, AND, and OR gates to create the logic for an XOR gate. It follows the Boolean expression: $Y = (\text{not}A \text{ AND } B) \text{ OR } (A \text{ AND not } B)$. The output is 1 if the inputs are different.

3. Full Adder Circuit

This combinational circuit is used for binary addition. It takes three inputs A, B and "Carry In" from a previous stage and produces two outputs: Sum and Carry Out. To implement the logic **Carry Out** = $(A \text{ AND } B) + (B \text{ AND Carry In}) + (A \text{ AND Carry In})$. At first the XOR gate was giving zero when we had all inputs 1 but then upon changing the behaviour, the full-adder started working fine

Lab Task 3: Implement and Verify the Following 4 to 1 mux in Logisim

For 4 to 1 multiplexer, 4 data inputs, 2 selection lines and 1 output is needed. The block diagram and circuit diagram is shown below.



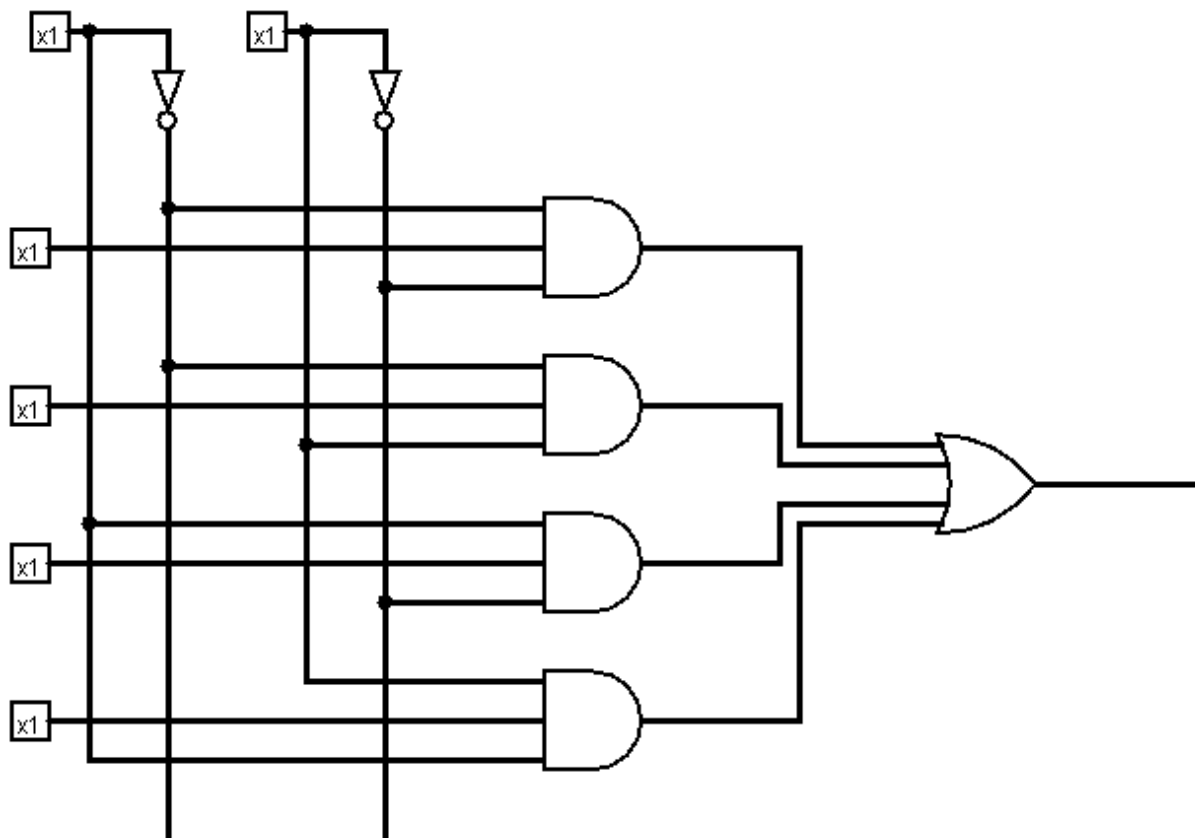
Block diagram and logic circuit of 4 to 1 mux

Enable (E)	Select (S ₁)	Select (S ₀)	Inputs				Output (Y)
			D ₀	D ₁	D ₂	D ₃	
0	x	x	x	x	x	x	x
1	0	0	1	x	x	x	D ₀
1	0	1	x	1	x	x	D ₁
1	1	0	x	x	1	x	D ₂
1	1	1	x	x	x	1	D ₃

Function table of 4 to 1 mux



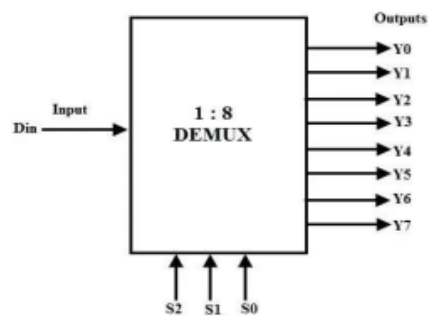
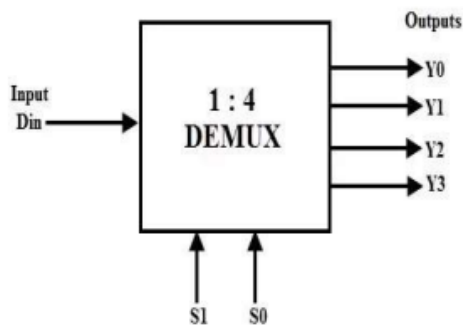
Solution:



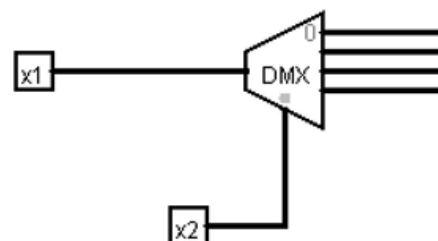
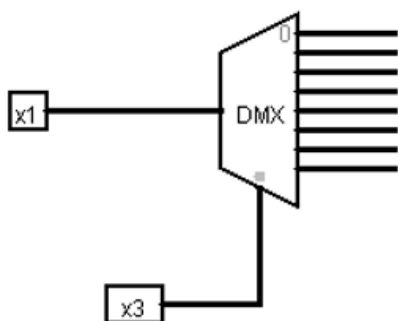
Explanation:

A 4x1 multiplexer was constructed from the scratch using the fundamental logic gates. A multiplexer chooses one of the 4 inputs (according to our combination of the switches) and outputs one of those lines. A multiplexer is useful for scenarios where we have multiple inputs for different cases and we need to cycle through each one of them if required.

Lab Task-4: Implement and Verify the Multiplexers and Demultiplexers using the built-in ICs in Logisim.



Solution:



Explanation:

1:4 and 1:8 Demultiplexers (demux) were experimented in Logisim. Built-in ICs were used for this; Demux essentially take one input and then give out multiple output lines depending on what was the input

Lab Task-5: Implement a Memory-Based System Using RAM and ROM ICs in Logisim

Components to Implement:

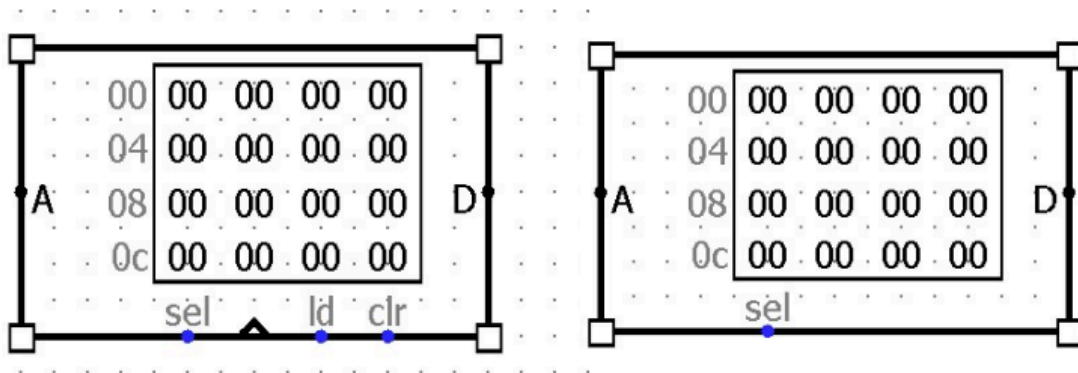
- **RAM (Random Access Memory):**
 - Implement reading and writing operations based on control signals.

ROM (Read-Only Memory):

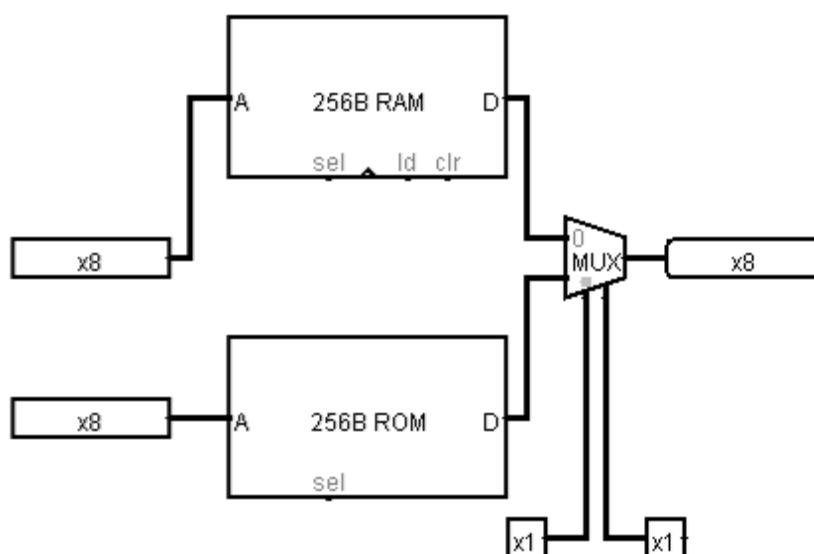
- Use the ROM to read data based on address inputs

Memory-Based System:

- Combine the ROM and RAM into a single system.
- Use a multiplexer to switch between ROM and RAM for data access.



Solution:



Explanation: Reading and writing operations were done in RAM modules based on control signals, allowing us to learn how they actually work. Data was read from ROM as we aren't able to write in ROM because it is read only memory. Multiplexer was used to switch

between ROM and RAM data accesses, we connected both outputs to multiplexer and then used switches to shift between either of the two lines